

ActInf Textbook Group ~ Cohort 4 ~ Meeting 5

(Chapter 2 part 1)

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hello everyone it's July 4th 2023 and we're in our first discussion on chapter two so let us head over there and before we jump into anything specific um does anyone have any General Reflections or or thoughts on chapter two just any any pieces that stuck with them from their reading or just overall reflections from chapter two on the low road yes if I could go I'm not sure whether yeah please um I was curious in terms of the equations uh that relate to free energy so the manifestations of free energy in those three or four equations mathematically how are those derived because in the textbook they kind of just seem to be axiomatically um placed there but I was wondering what is the underlying mathematical root of those equations from which we work thanks Ollie thank you yeah the derivations are the four equations at least 2.5 and uh 2.6 is um I mean we've done basically uh complete steps for the derivations of each of those equations and I'll upload a file explaining each step after the meeting if you want but I'm not sure where to put the files uh Daniel could you please uh I mean where do you think would be most helpful to put those files file format is it uh it's a PDF so what one option is to just at the top of the notes you could always just drag in a PDF or I just drop it here another option is to um in the GitHub repo for the textbook to add it there in some format um but yeah certainly dropping a PDF or taking screenshots in here would be awesome but yeah this is this is a great question and one more thing about the derivations of those Central equations is uh they're not necessarily uh evidence from the textbook or the way they're expressed in the textbook how we can get from each line of the equations to the next line so I agree we need some intermediary steps and even we need to employ some mathematical um some mathematical theorems theorems and rules such as Jane's inequality rule to get to some of those equations but I hope I mean the derivations we've worked out would be helpful if people want to follow the exact details of each derivation yeah that's awesome work with um Ali and Jonathan and Jacob also did some of these preliminary um connections for a few of the equations cool oh and we're gonna come to to these equations and we'll talk about them a lot

more soon but any other just overall thoughts or or reflections at the something that was interesting to me is there's kind of a however loose identification between an agent an active inference and their priors this is pulling from page 39. that we can say in active inference the identity of an agent is isomorphic with its priors or the relationship between I suppose the the agent and their their preferences yeah the priors are what the thing brings through the needle of the present to confront the incoming observation to then be updated into the posterior but then that just becomes the prior for the next moment so like if it's not coming with you through the priors it's not making it through the bottleneck of the presence it has been forgotten so everything has to come through the presence in order for the past to influence the future that's like the markovian property of like a Markov chain so that Markov blanket necessarily but actually that is a markup blanket through time past only influences the future through the present so everything has to flow through the present in order to have influence but that is a very interesting claim any other um thought thoughts or ideas on two just overall cool we'll we'll look at some questions but just the big picture for the next month I guess for all of July two weeks on chapter two that's the low road and then two weeks on chapter three that's gonna be the high road so first we're going to talk about how and this is going to get us from the basic tautologically true Bayesian theorem all the way on through heuristics that are used to fit Bayesian models of perception and action that's going to get us to active inference and then we're going to kind of like take a breath reset start not from the how on the low road but the why on the high road and that's going to get us to the same place and then that will drop us off at chapter four which is going to be like another level of depth into the generative model that is used at active inference um first I'll just check Okay cool so I'm going to chapter two um questions for cohort four you can just follow the little um token if you want okay all right why does the minimization of Divergence not also lead to the maximization of model evidence are you not generating new accurate priors that are evidenced by the incoming sensorium so if the person who asked this is here they can unpack a little bit or anyone else can provide a thought or an idea on this yeah so I wrote this just uh with an eye on the equation um let me be good to get it after possible because no worries if not um which has the kind of action perception dual aspect so the first bit is perception with the KO Divergence and the second verse with action my question therefore is if we are getting closer to Optimal Bayesian inference would you lowering the bound on free energy surely optimizing or getting close to Optimal Bayesian inferences therefore driving us to update our priors in a way that would be the way that basically perfectly reduces prediction error and therefore minimizes

free energy so I wasn't sure whether is it just that in reality action and perception are always coupled or is there uh is there something you know is there a limit to optimizing perceptual inference yeah great um topics which equation were you referencing just so we can look at it so it's so if you go back on the textbook um it's worth so it's on 2.5 and that it's the last one I've uh so if you keep going I'm sorry 2.6 um yeah so it's the Divergence evidence one yeah yeah okay okay okay great all right equation 2.5 variational free energy okay so first just to just to give a thought again anyone can raise their hand up if they if they want to add more the minimization of Divergence does lead to the maximization of model evidence so the the um um relationship to keep in mind is that the model with the most evidence for it is going to be the one that minimizes surprise minimization of surprise is itself intractable and so we use this bounding approximation which is the expected for energy so let's think about this with the heights of the children in the classroom so we want to have the maximum evidence estimation for the heights of the children in the classroom that's like the ultimate statistical problem that we want to solve now if we had the model parameterization that minimized our surprise so as new heights were drawn we were minimally surprised that would be equivalent to having the maximum evidence model and then surprise itself which is just like the log of P that is intractable and that's why we have a bounding function so the minimization of Divergence is used um as a tractable way to because this is going to be a constant with respect to Q so in terms of the the distribution that we actually control the knobs on Q this is not going to have um any gradient to it and so that's why variational free energy can be incrementally optimized by reducing the KL Divergence here [Music] and you are generating priors that are uh you're you're updating priors based upon incoming sensory data I'll just note though that action is not included in this equation this is purely a functional so it's a function of a function it's a functional variational free energy of Q the distribution that we control variational approximator and the data y but there's no action selection here anyone else want to add a thought on um yeah Ali thanks well actually uh this Duality between the minimization of the free energy on one hand and maximizing the evidence afforded by the generative model and the other is what usually referred to as uh self-evidencing mechanism after Jakob Howie's well-known paper from 2014 or 16 I guess so uh the idea is that uh well I mean a particle and by particle I mean The Joint set of internal and blanket states of a system uh will look as if it is gathering evidence for its own generative model and where its generative model also entails the prior belief that uh that it um it's somehow acts upon upon later by choosing proper uh sets of uh I mean policy and by policy I mean a set of actions uh selected by um the agent or the

particle as somehow minimizing the expected free energy at this time so variation of free energy provides a kind of a kind of upper limit to uh to to the way that uh the agent tries to minimize its perceptual uncertainties but then later on in order to effectuate that uncertainties and to uh kind of have causal intervention onto the environment then it needs to select a policy in order to minimize its expected free energy so in this sense both variational free energy unexpected free energy can be seen as parallel or dual to each other and we can obviously see from the form of the equations of 2.5 and 2.6 that they're actually mathematically and formally are parallel thanks another way to to um see this is like we're gonna come back to it um multiple times but variational free energy is like the real-time perceptual how's it going because it is purely inferential it's real time as a functional of beliefs and incoming data so if we're talking about body temperature and homeostasis this is just like how the body temperature is and so we kind of latch our model's performance and the the desire to not under or over fit that is latched to the ultimate observable layer with incoming data but this is a passive equation but this this is the same variational free energy that you would see in a variational autoencoder so if you had a some Bayesian statistics problem and you're just trying to um interpolate um pixels in a variational autoencoder you wouldn't want to over under fit this is exactly what you would have and then in order to have causal efficacy and the imperative there just like mentioned this is going to be a constant term with respect to Q so we can minimize the Divergence this KL Divergence which is always bigger than zero and that pushes down F towards surprise if we had the minimally surprising model we'd have the model with the most and then in 2.6 we get into causal efficacy in a sort of prospective way Young that's just a I think it's a minor issue but because because our ZIP objective is to lower the variation of free energy it seems to me like it's better to explain surprise as the lower bound of variational free energy rather than seeing that the variation of the energy is the upper bounds because well the our objective is to reduce these cable Librarians of 2x with the posterior so that the free energy itself can be minimized to the surprise itself so I think it's more intuitive to explain as surprise being the lower bound of the variational free energy yeah that's that's a good point and also once you start doing oh well there's negative f and and so the upper and the lower I I I think whichever way you you want to approach this um this Divergence from when the KL Divergence is minimized then the distance between these two distributions is lower sorry not the distance the Divergence the only difference between a Divergence and a distance in this setting is like a distance is the same from A to B as B to a and it turns out the KL doesn't have that property which is why it's called the diversion it's not

a distance um however when variational free energy is minimized then the approximating belief function Q with respect to the incoming data are doing as well as they can at bounding surprise that's the best fitting model with the most evidence I mean if you knew optimally how surprised to be by uh what was going to happen next in the movie you would have the best model of what was about to happen in the movie and so maximum model evidence can be approached by minimizing surprise minimizing surprise is itself intractable compute so we minimize around on surprise which is the variational free energy Ali actually that's uh exactly the elbow or evidence lower bound as encountered in RL literature or in um I don't know some other resources but yeah I agree that sometimes it can at times it can be confusing because uh elbow is basically the negative variation of free energy right so um yeah that's exactly almost exactly the same concept yeah and then um uh Martin you asked the chat can surprise be lowered through action well yes these are the two ways to to minimize free change your mind perception learning change the world through Action Now action cannot directly reduce your surprise about sensory outcomes so let's just think about body temperature we expect and prefer 37. we're getting surprised because we're at 40. now no action is going to pick up the temperature and move it back to 37. but in action May influence the world turning on the air conditioner or taking off the jacket so that future sensory observations are minimally surprising and that's exactly what equation 2.6 describes which is we're going to evaluate future courses of action turn on the air conditioner or don't these are our two policy options evaluate these two alternative policies in terms of two different criteria what we stand to learn here y tilde is the only difference between these two double lines the KL Divergence that's what you're taking the Divergence between so if y tilde gave you no information then this KL Divergence would be zero so if the policy doesn't bring you any if the new observations don't bring you information the informational gain is whereas if you stand to learn a lot from the observations that you expect to get under that policy the the informational gain or the epistemic value is high pragmatic value is about the alignment of observations with your preferences and so actions are selected prospectively in order to um balance the price but that also includes this epistemic components so it's not just like the kind of dark room well then why don't we just close our eyes and you know curl up because Things become more surprising while you're not looking so informational gain is still part of the picture but yes absolutely and that's why it's self-evidizing is this kind of like unifying imperative that helps us understand the real-time flow of belief updating and sensory perception on one hand and extends that prospectively into the actions to let action selection case with expected free

energy so can surprise we lowered their action not directly however actions are selected prospectively based on their causal efficacy in the world with the imperative of bounding surprise about those observations Darius yes I recognize um I may be jumping the gun on this and maybe we will come to it but um in order to select uh perspective action policies so as to minimize future free energy presumably is to run the sort of causal regression back there has to be some trials um regarding the probability of that policy reducing free energy so my question is according to the active inference framework what are the initial priors that govern um action and I'm talking even here at a very sort of Developmental level um I imagine that genetics has a role to play but is there also just a picking out of good outcomes initially on basically randomness yeah yes this is a very a very uh good question and it speaks like some of the the challenging aspects of modeling which is what do you explicitly versus implicitly bake into the model so this is like looking a little bit ahead to figure seven three it's the same as figure 4.3 basically the um causal efficacies in the world are captured by the B Matrix if I do policy one versus policy two how is that going to change hidden States and then hidden States emit observations so how do Agents come to know that turning on the air conditioner reduces the temperature or that putting on jacket increases the temperature so that's the question of of the B Matrix Being Fit obviously if your B Matrix were washed out action would be meaningless like if you didn't if if you didn't know or if your actions didn't have any consequence then action selection is kind of for naught and then depending on on the situation you could think about B being inherited like inherited prior or like the first prior potentially developmentally also B can be lurked so the consequences of action can be learned from an initially plastic B Ollie actually this is a question that's been asked time I mean over and over again uh has kind of criticism for active inference framework regarding how exactly those prior beliefs um emerge from so um I mean uh talking a bit more uh theoretically philosophically or even more fundamentally uh well fep starts from the priest's opposition that there is an agent or a particle with a certain internal States and then attempts to describe the geometry of the information uh that's been related to the interaction between those internal States and the external States through Markov blanket so fep by itself doesn't say anything about how those internal States emerge from at the first place of course it provides an account for the self-organization through I mean describing uh describing the process that such internal states can increasingly get more complicated becoming higher dimensional and so on but I mean by itself it doesn't provide any account uh for any explanation for the emergence of those uh internal states of the first place so if we take that as uh the basic premise of the active

inference I mean as a precondition for the application of fep then I think this question or this criticism can be resolved uh because obviously fpp never attempts to provide anything more than it claims right so yeah that's uh I think one of the crucial points that we need to keep in mind uh regarding what exactly the are the claims of are the theoretical and empirical claims of FTP and active improvements awesome and this this reminds me of the uh delightful actual constants um argument the free energy principle it's not about what it takes it's about what took you there if you're ready for some um philosophical tongue twisters the free energy principle is not concerned with the conditions of existence so not what has to be the case for something to exist that's an ontological analysis rather with what must have been the case given that a thing exists so the presupposition of there being a thing is what Ali highlighted it's not about figuring out what it takes to be alive again what conditions lead to life flip it it's about a system of interest and understanding what must have happened for that system of interest to have been that way which is self-evidently behavior so it's not that self-evidencing behavior begets life necessarily although that also may be true it's that things that are persistent whether they're inert and simple or conservative um active systems or even strange sophisticated cognitive systems whatever level of complexity the thing is we're going to be able to do this kind of fep or active inference analysis but that doesn't mean that this is a forward-looking analysis about what it takes to be alive but this is an excellent work by Axel okay cool let's continue through these questions but these are these are fun page 22 the results of inference are not necessarily accurate in any objective sense for a number of reasons biological creatures operate on the basis of limited computational and energetic resources ultimately as all computers do too with these limitations influence the strength of priors in addition to confidence and thus minimize Bayesian surprise for example observations might give rise to surprise but the generative model is not updated uh due to the inferred energetic or cognitive costs of updating practical examples being not wanting to change an opinion because doing so requires investment of time uh to thank discern Etc yeah I think it's it's a great Insight people can feel free to add however strategy selection cognitive strategy selection may or may not have ultimate accuracy as it's imperative as the question highlights whether it's a energy ultimately energetic cost or it's some type of like psychological or energy or cognitive cost of updating or even a Time cost or an uncertainty cost or a social penalty type cost there are a whole host of reasons that an update could be considered Bayesian optimal and not fully or only align with the most accurate account Darius just a question what hope to mine was might that not be the minimization of prediction error or free energy at a higher level

of the hierarchy so I may not want to change my mind because of social pressures in turn fulfilling my prediction that I will not be socially ostracized so will a Bayesian with an optimal Bayesian inference engine might this still be optimal it's just that we're getting skewed by looking maybe not holistically at the the entire hierarchy yes this is exactly exactly it like how how can um um even pathological or or harmful Behavior be understood as Bayes optimal and and the answer there is given the setup of the generative model the prior is the way they are and the state um transition inference is the way they are the attention the way it is then name your DSM condition it is a Bayes optimal given the totality of the cognitive model and so it's it's not like some behavior is is um like let's just say rational or irrational they're all doing optimal Bayesian inference the ball is rolling downhill now you might not like where the ball rolls or Society may say that the ball should have gone this way not that way it's all like water flowing to the bottom of the hill or the ball rolling to the bottom of the hill on the landscape and that's optimal Bayesian inference so that's that's a totally great on point and they'll talk more about like optimalities measured in relation to expected free energy well if you have a distorted perception of how things are happening in the world then that agent's optimal selection again doesn't mean that the payoff is going to be the highest it just means that it's it's a optimality concept that's um agentic so optimal Bayesian inference doesn't mean best of all possible worlds all the agent does is when you know Fitness is optimized the game theory is transcended optimal basing inference just means given the generative model the ball rolls down the hill and so if you account for for example higher order considerations then you you can end up with what might otherwise be seen as like aberrant or pathological Behavior so for example um uh uh being unable to um uh confidently predict the reactions of others in a social setting might lead one to pursue um a sort of um self-oriented Behavior because it has understandable predictable consequences like self-stimulation again that's not to say that that it's optimal in terms of social reward received it's just under the optimality Criterion that generative models are actually playing out under all neurodiversity Beyond humans even just cognitive diversity is being modeled with this optimal Bayesian framework but connecting back to the um cybernetics good regulator theorem and requisite diversity while it's true the results of inference are not necessarily accurate over the developmental or evolutionary time scales at least the results of inference have to be adequate enough to keep you in the game otherwise you're very quickly going to be swept off the table so that's kind of this qualified pragmatism scientific realism like it it's not not trying to paint how it really is out there but causal architectures of the external World though they're not directly observable have to be at

least understood well enough in order to make adaptive decision making in a dissipative or adversarial niche well actually um what you described accounts for a pathology without Insight when the patient suffers and knows something is off I am not sure it's described this way pathology with Insight present depression yeah meta cognition okay prediction error surprise variational free energy these terms are related to one another but sometimes the relations are not immediately clear causing some confusion the common objective of perception action can be formalized as a minimization of the discrepancy sometimes this is operationalized in terms of prediction error when is the discrepancy not operationalized as a if we calculate surprise as per page 19 is this a different um than the calculation for related terms what is the possible confusion in interchangeably using these terms okay I'll give a thought but but I'll hear anyone else please feel free is like the simple differential or the simple discrepancy I thought it was going to be 10 it came in at 11 so the the simple discrepancy was this amount of units of temperature thought it was going to be 10 degrees came in at 11. the prediction error was plus one unit of temperature in contrast surprise is an information theoretic quantity it's measured in Nats or in bits but surprise is like looking at the distribution how surprising is that given observation kind of like a z-score in statistics like oh this was three standard deviations now it may be the case that one degree temperature is one standard deviation and so it was one standard deviation a surprise for that plus one prediction error and then um shown in equation 2.5 they're related to each other because it where the prediction error is surprise is minimized but not necessarily zero they're still non-zero surprise associated with picking the exact highest point on the gaussian minimized surprise but not zero surprise and similarly picking that exact midpoint on the gaussian would minimize variational free energy okay I thought that the classroom was four feet plus or minus one I pull a height it's four feet so it's a minimal surprise minimizing minimizing prediction error so they're kind of all um colleagues and helping us point to to different aspects of model fit but they're not the exact same quantity Ali and then anyone else also it relates to the difference between the exact Bayesian inference and the approximate Bayesian inference because uh basically when we talk about it refers to the exact amount of the surprisal that needs to be minimized but when we talk about variation of free we uh we utilize the approximate Bayesian inference techniques in order to come up with a bound or more precisely an upper bound to minimize the surprise so it's not the exact Bayesian inference because it would be track intractable and most real world situations but rather it provides an alternative way of dealing with the complex situations by providing an upper Bound for the surprisal namely the variational free energy uh also uh I

think it might I mean it might be helpful to point out uh the paper by alirezamodu shanichi which provides 14 different definitions for the surprisal in different contexts so we also had a guest stream with aneurizoma dushanichi which I recommend watching and looking up that paper as well yeah this was very um very rich paper and stream and discussion where they derived I think yeah how many 18 or 14. surprise Concepts included and show you know again just intuitively if your model is fitting as well as possible if the evidence is the highest you'd be right to expect that surprise would be lowest in general but it turns out that there's actually a plethora of ways that surprise is computed and they're they're not exactly equivalent and and we expect that many other ontology terms attention anticipation all these kinds of terms where we have a broad everyday sense of the term and then we also make technical definitions of them in the active inference ontology many other terms may have also um just like words in the dictionary have multiple bylights great question though equation 2.6 so the trade-off between Information Gain let's copy it in you trade off between Information Gain and pragmatic value is quite understandable so that's referring to this kind of Top Line classical decomposition of expected this is our unified imperative for Action selection which is we should select policies based upon their pragmatic and epistemic value not to talk about the other guys but in reward Centric learning you need to coerce epistemic value into pragmatic value because it's like a single currency of of pragmatism or reward so you have reward itself and he said well it's it's worthwhile to go on this little exploration because the expected reward or the expected utility is such and such so I I would you rather have one dollar on the ground now or ten percent chance of a hundred by going on this exploration and so reward learning would go on that um route depending on some risk discounting which we'll we'll look at whereas an expected free energy we're just jointly considering both the epistemic and the pragmatic now we're going to look at some different decompositions of G how expected ambiguity and risk interact or are counterbalanced is quite hard to be understood can anyone provide a good explanation so does anyone want to give a thought on how would they read the second line and I think we'll all look forward to the derivations and also look to improve the the natural language descriptions with just like the verbatim readings um but what what does anyone see in this second line I'll I'll just give a thought so H is entropy so this is how blurry or sharp a distribution is is captured by its entropy the first term expected ambiguity here we have a distribution of observations through a Time y tilde given hidden states to talk so this is like how entropic how um um sharply defined will observations be so you can imagine a super if there's two outcomes it's going to be really ambiguous if both the outcomes have this have

um messy observations you know whether we win or lose if we're getting this static on the radio then the ambiguity is high whereas this term is going to be um favoring policies that lead to unambiguous resolution of the relationship between hidden States and sensory data so here the kind of sub imperative is pursue policies big fancy e expectation pursue policies that have um less ambiguous expected entropies this so so this is just sensory this is only about the mapping between the hidden States or time and the sensory observations um that are happening through time the second term we have the KL Divergence double line on the left side of the parentheses are the sensory observations Through Time and so it's the difference between the sensory observations condition on what we do and the sensory observations conditioned on what we prefer C this is risk not in the danger sense specifically but rather a risky outcome or a risky policy is one that leads to um a range of outcomes a non-risky policy is one that Maps tightly to a certain outcome so this expected ambiguity term helps us um pursue policies that give unambiguous sensory data about hidden states of the world and this risk term helps us pursue policies that are causal intervention has efficacy in the uh um being able to shape sensory outcomes this next yeah Andrew uh I just wanted to say it the the most health prior to your explanation for understanding these terms were just the simple examples given in textbooks so so with ambiguity they give the example of like the probability of heads or tails in the coin flip conditioned on whether it is sunny or raining will be maximally ambiguous is there's no relationship between the weather and the 50 50 chance of heads or tails so um it seems to relate to something like independence of events uh in that case I mean it's entirely independent you know the result of your coin flip and um very distinct and it doesn't provide you any useful information about the other scenario um and then with with risk um similarly to what you were saying uh the kind of one-to-many mapping between policies and outcomes for example a highly risky policy uh or action is related to uh playing a slot machine you know or say you have a slot machine that has like three potential outcomes each with equivalent you know probability is oh I do this one action but there are a variety of outcomes you know that could come out of this and so there's very high risk uh to your action as opposed to you know um holding a slot machine and having a guaranteed win every time nice awesome uh I actually wrote down this question I got a bit confused about um well so policies that seem to be uh reducing the ambiguity also seems to reduce the risks so like it made me kind of confused because well both both values would be would be heading towards the same policy so it doesn't seem like it's a trade-off at all um compared to the information gain and pragmatic value which seems to be very intuitively and create a trade-off uh I

couldn't catch the um uh a good example or a good explanation about how these two values expected ambiguity and risks are interacting with each other that's a very very good point um although explore exploit is kind of like a classical trade-off and it's one that's going to be approached in the book like um you could imagine situations where there's two policies and one of them gives more epistemic value and pragmatic value and similarly you could imagine situations where they are um in trade-off or attention and so similarly it would be helpful to think of examples like real or contrived um where ambiguity and risk are aligned or not Ollie what do you think you're immediately or I'm sorry oh yeah can you hear me now uh all right so the expected free energy somehow unifies uh various accounts of um resolving the ambiguity from different uh disciplines and different um I mean from several with Townsend psychology machine learning and so on so for instance uh accounting for the risk both in terms of outcomes and in terms of States uh is inspired from a common and inversky's work on the risk sensitive policies as applied to economics and also the Callback Library control engineering so here uh actually minimizing the risk somehow course pounds it corresponds to aligning our predictions to our prior preferences but well in the absence of um the risk all we have is just the ambiguity uh that uh that that is basically accounted for in the accounts from uh James and also Lindley uh in which uh they they only account for the resolving of the uncertainties through uh the uh through diminishing the ambiguities and not accounting for the risk implied in each situation uh so uh I think we're short on time but it can be explained uh in a bit more detail how exactly uh those uh three or four lines of equations can be related to each other and uh how each of them corresponds with some of the other accounts of uh how how we can resolve those ambiguities and uncertainties through different techniques but nonetheless expected free energy uh to the best of my knowledge is I mean provides uh the most all-encompassing and unified way of accounting for all of these um various techniques of reducing the uncertainty but we can talk about that a bit more in the next session I guess yeah thank you Ali figure 2.6 summarizes a few of those threads it shows how like in the absence of what you're left with is an information seeking imperative conversely in the absence of epistemic gain you're left with naked epistemic utility Theory and under another restricted subset kind of special case of G you end up with risk sensitive policy which is part of that common inversky um line of work um well glad that we were able to get to the uh um cohort 4 chapter 2 questions uh next week we will be a different time but we'll continue on chapter two so whether you'll be able to be here or not please feel free to add more questions in and and uh upvote other questions I mean there's almost 50 questions but please

please add up votes and we'll uh improve the discourse and maybe record some short videos
and and continue these great topics as we proceed along the low road so thank you everybody
see you next time bye thank you thank you